PickleVerse Test Plan and Evidence Register

Report 5 · Proposed cases and current verification limits

Working draft for team and reviewer discussion. Source snapshot c32f2a4c0a54 on 23 October 2026. No game/backend execution or code edits were performed. PKV is a working abbreviation; official code, names and approvals remain unconfirmed.

This is a test plan and a traceable case register. Every game/backend case is NOT RUN in this task because the authorized work is documentation-only. Static documentation checks are reported separately and are not counted as passing product tests.

1 Test inventory and counting

Source	Count	Rule	Subtotal
Mandatory features	18	3 per feature	54
Main workflows	6	1 E2E per workflow	6
Declared state and rejection transitions	118	1 per transition	118
NFR groups	6	1 measurement case per group	6
Proposed total	184	Unique IDs; no double counting	184

Automation target after agreement: at least 92 cases actually run in CI. Actual CI executions verified here: none. Overall passing target: at least 166 of 184, plus every mandatory main-path case passing. Main demonstrations: 6 of 6; exception demonstrations: at least 5 of 6. No open Critical/High defects are permitted at Defense 1.1. The proposed numbers are not yet signed commitments.

2 Required setup and evidence

Use a declared staging environment with API/schema/client versions, isolated seeded users and deterministic content. Seed two human accounts for 1v1 and four for 2v2, an Admin and a non-Admin, an unverified account, locked/completed drills, eligible/already-claimed rewards, tied Elo records, a moderation report and authorized/foreign replay samples. Use synthetic identities; never include live passwords or signing keys in the repository.

Seed_Manifest.csv defines the planned fixture keys, Elo 1500 matched accounts, team membership, tutorial/reward states, legal 11–9 and 11–8 match scores and analytics sample counts. It is a specification only: no accounts, service resources or database rows were created. Record generated UUIDs and the actual catalog version in protected staging execution evidence.

Each execution records TC ID, commit, environment, seed version, date, device/build, steps/data, expected and actual result, pass/fail, log or screenshot link and responsible named member. Keep API contract tests separate from rendered UI and physical-touch performance tests. Galaxy A32 simulation cannot establish real-device FPS or latency; iOS acceptance requires a real signed build/device.

3 Main and exception demonstrations

WF01 Register learn and complete a first drill

Main Submit and verify account; Open tutorial and perform controls; Choose an unlocked drill; Start server attempt and submit measurements; Complete and refresh stars and first reward. A verified player completes a first drill and sees the persisted result.

Exception Expired OTP or invalid trial data is rejected; account/attempt can be retried without duplicate reward.

Presenter slot M1; staging seeded setup; actual run NOT RUN.

WF02 Practice against an adaptive bot

Main Choose bot difficulty and court; Run entry cinematic; Play serve and rally with legal inputs; Resolve points and match end; Wait for end notice then show player result and statistics. The player receives a legal match result and coaching evidence.

Exception Illegal kitchen/serve action is penalized; an aborted/unverified result cannot award a duplicate reward.

Presenter slot M2; staging seeded setup; actual run NOT RUN.

WF03 Find accept and finish an online match

Main Join eligible skill queue; Poll canonical offer and accept; Wait for all required human acceptances; Start Photon Shared session and play; Report consistent results and read finalized rating. A canonical match result updates each participant once.

Exception A decline/timeout cancels the offer; a conflicting report blocks normal finalization and applies server policy.

Presenter slot M3; staging seeded setup; actual run NOT RUN.

WF04 Create a private team match

Main Find a friend or share room code; Create selected room mode; Join and assign valid teams; Confirm ready roster and start; Finish and inspect authorized statistics. A coded room starts with the declared 1v1 or 2v2 roster.

Exception Wrong/full room or unauthorized team/start command is rejected; the room stays consistent.

Presenter slot M3; staging seeded setup; actual run NOT RUN.

WF05 Review progress and claim rewards

Main Refresh server progress; Inspect stars and error aggregates; Open suggested drill or replay; Claim eligible mission/achievement reward; Refresh wallet and continuation target. The player sees next practice, history and exactly-once rewards.

Exception A repeated claim does not credit again; offline practice/sync is a required gap and must be demonstrated after implementation.

Presenter slot M1; staging seeded setup; actual run NOT RUN.

WF06 Administer users content and operations

Main Authenticate as administrator; Inspect a user or moderation report; Apply valid access/status decision; Create and publish playable lesson configuration; Inspect period-defined operational analytics. Authorized actions change service data and support operational decisions.

Exception A non-admin request is denied; invalid publication/status data leaves existing records unchanged.

Presenter slot M4; staging seeded setup; actual run NOT RUN.

4 Test distribution and ownership

Workflow	Owner slot	Feature	E2E	Transition	NFR	Total
WF01	M1	9	1	29	0	39

Workflow	Owner slot	Feature	E2E	Transition	NFR	Total
WF02	M2	6	1	14	2	23
WF03	M3	9	1	34	1	45
WF04	M3	6	1	13	0	20
WF05	M1	12	1	10	1	24
WF06	M4	12	1	18	2	33

Role slot	Unique cases	Workflows	Named confirmation
M1	63	WF01, WF05	Pending team assignment
M2	23	WF02	Pending team assignment
M3	65	WF03, WF04	Pending team assignment
M4	33	WF06	Pending team assignment

5 Full register and execution method

Test_Case_Register.csv contains all 184 unique cases with feature/workflow/owner, preconditions, data, steps, expected result, execution method and actual status. Feature_Traceability.csv links each feature to use cases and tests. Rejection/self transitions are explicit cases and do not inflate main-flow pass counts.

Case example	Purpose	Expected result
TC-F01-01	Register PLAYER_NEW; verify valid OTP; log in; complete the required self-assessment target; load own UUID/profile; refresh token; log out and reject the revoked refresh token.	Create and verify an account; sign in and load the same server profile; reject invalid credentials; record self-assessment for initial placement.
TC-F01-02	Submit invalid/duplicate inputs for Account and onboarding	Reject invalid mutation and preserve prior state; no duplicate reward/result
TC-F01-03	Repeat using an unauthorized account or unavailable dependency; retry safely	Show actionable error; no foreign data access or partial committed mutation

6 Performance and reliability measurement

N01 Interactive performance

Median and p95 frame time; input-to-visible feedback; technique feedback delay. Proposed: median ≤16.7 ms and p95 ≤25 ms on an agreed midrange device; input p95 ≤100 ms; coaching <1 s. The proposal states stable 60 FPS; any relaxed percentile needs approval.

Release builds on Windows and Galaxy A32; 15 minute day/night sessions in all four play scenes; capture CPU/GPU/frame-time and touch traces.

N02 Network responsiveness and capacity

RTT; concurrent players and matches; matchmaking errors. Proposal: typical VN RTT <100 ms; 2,000 players / 500 concurrent matches; no cross-match membership or duplicate result.

Seeded staging multi-client load and network impairment; measure p95 RTT separately from server command latency.

N03 Availability and recovery

Monthly successful service availability; tested backup/rollback. Proposal: ≥99.5% monthly availability; proposed recovery targets RPO ≤24 h / RTO ≤4 h require approval.

External probes plus service metrics; restore a staging backup and deploy/rollback a known version.

N04 Platform and usable controls

Acceptance pass matrix; safe-area clipping; target hit area. Windows 10+, Android 9+, iOS 14+ as proposed; all main flows pass; no clipped essential controls; proposed touch targets ≥48 dp.

Physical Windows/Android/iOS test matrix and landscape safe-area screenshots; simulation does not prove physical performance.

N05 Security and trusted data

Unauthorized actions; replayed commands; transport/at-rest protection; simulation authority. JWT access control; reject all unauthorized mutation/replayed conflicting commands; proposal AES-256 and dedicated authority require deployment/design evidence.

Auth/RBAC negative tests, duplicate/race tests, TLS/secret/backup configuration inspection and malicious-peer simulation.

N06 Offline data and learning quality

Offline completion/sync duplication; classifier and recommendation quality. Proposal offline training plus exactly-once permitted sync; proposed ML macro-F1 ≥0.80 and recommendation acceptance ≥80% require labelled evaluation and agreement.

Disconnect/reconnect drill tests; account-switch conflicts; held-out labelled dataset with confusion matrix and recommendation review.

7 Risk priorities and exit criteria

First repair and contract-test admin login, SQL status values and playable publication. Then verify online accept/cancel/report/disconnect races with multiple clients, exactly-once reward retries and replay ownership. Dedicated simulation/offline/ML are proposal gaps requiring implementation or an explicitly approved scope/design decision. Freeze the agreed documents and code at review-2 only after team confirmation, then rerun the unmodified standard self-review prompt.